6. Write a Servlet to demonstrate session tracking using HttpSession. Implement a simple login system where the user's session is tracked.

AIM

To create a web application that demonstrates session tracking using the `HttpSession` interface.

ALGORITHM

1. **Input:** The user enters their credentials on the login page.
2. **Session Creation:**
   - Upon successful authentication, the Servlet creates a session using `request.getSession()`.
   - The user's identity is stored as a session attribute (e.g., `session.setAttribute("user", name)`).
3. **Session Tracking:**
   - When the user navigates to the Dashboard, the Servlet checks if a valid session exists.
   - If `session.getAttribute("user")` is not null, the dashboard is displayed; otherwise, the user is redirected to the login page.
4. **Session Termination:** A "Logout" button is provided to invalidate the session using `session.invalidate()`, ensuring the user is logged out.
5. **Deployment:** Compile the Servlet, map it in the `web.xml` or via annotations, and deploy it to Apache Tomcat.

SOURCE CODE

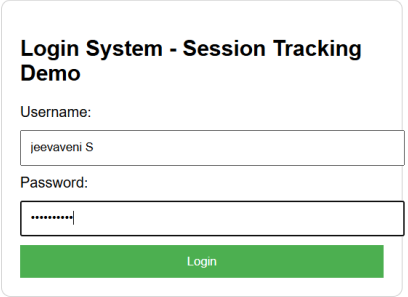
HTML File

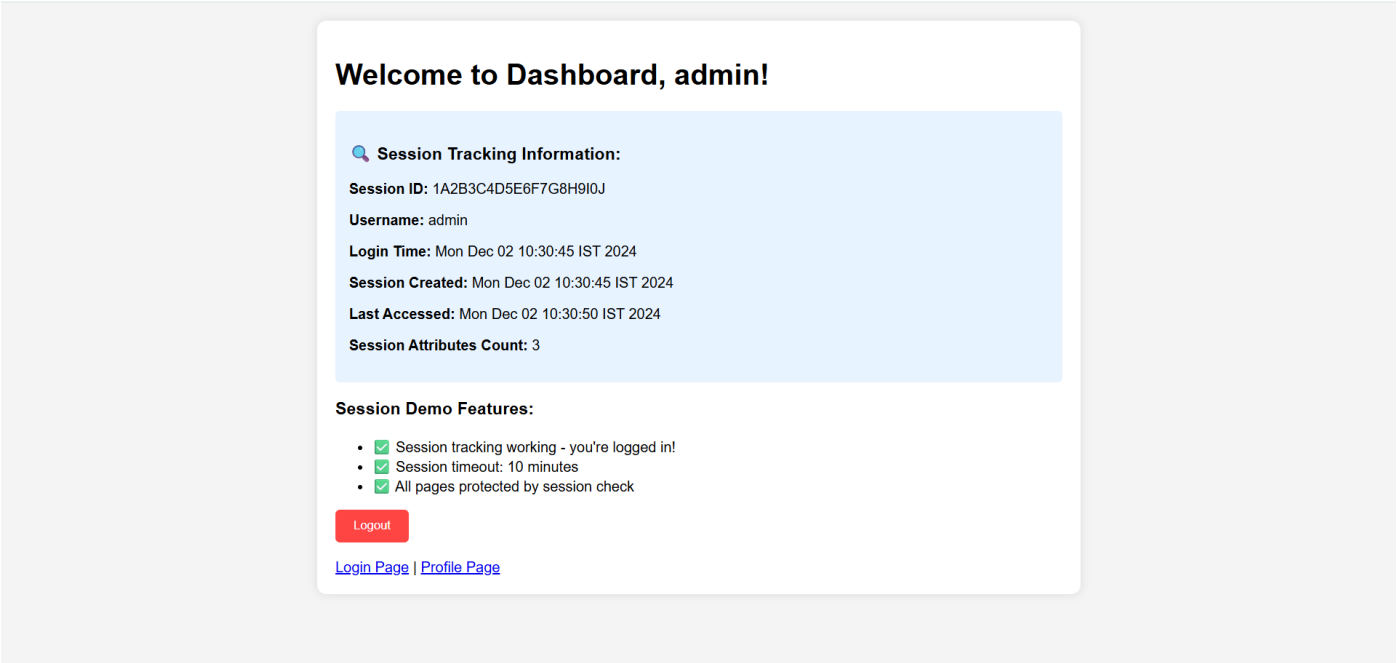
```
<!DOCTYPE html>
<html>
<head>
<title>Session Tracking - Login</title>
<style>
body { font-family: Arial; margin: 50px; }
.container { max-width: 400px; margin: auto; padding: 20px; border: 1px solid #ccc;
border-radius: 10px; }
input[type=text], input[type=password] { width: 100%; padding: 10px; margin: 10px 0; }
input[type=submit] { background: #4CAF50; color: white; padding: 10px; border: none;
width: 100%; cursor: pointer; }
</style>
</head>
<body>
<div class='container'>
<h2>Login System - Session Tracking Demo</h2>
<form method='post' action='login'>
<label>Username:</label><br>
<input type='text' name='username' required><br>
<label>Password:</label><br>
<input type='password' name='password' required><br>
<input type='submit' value='Login'>
</form>
</div>
</body>
</html>
```

Dashboard Page

```
<!DOCTYPE html>
<html>
<head>
<title>Dashboard - Session Demo</title>
<style>
body { font-family: Arial; margin: 20px; background: #f4f4f4; }
.dashboard { max-width: 800px; margin: auto; padding: 20px; background: white; border-
radius: 10px; box-shadow: 0 0 10px rgba(0,0,0,0.1); }
.session-info { background: #e7f3ff; padding: 15px; margin: 10px 0; border-radius: 5px;
}
button { background: #ff4444; color: white; padding: 10px 20px; border: none; border-
radius: 5px; cursor: pointer; }
</style>
</head>
<body>
<div class='dashboard'>
<h1>Welcome to Dashboard, admin!</h1>

<div class='session-info'>
<h3>🔍 Session Tracking Information:</h3>
<p><strong>Session ID:</strong> 1A2B3C4D5E6F7G8H9I0J</p>
<p><strong>Username:</strong> admin</p>
<p><strong>Login Time:</strong> Mon Dec 02 10:30:45 IST 2024</p>
<p><strong>Session Created:</strong> Mon Dec 02 10:30:45 IST 2024</p>
<p><strong>Last Accessed:</strong> Mon Dec 02 10:30:50 IST 2024</p>
<p><strong>Session Attributes Count:</strong> 3</p>
</div>

<h3>Session Demo Features:</h3>
<ul>
<li>✅ Session tracking working - you're logged in!</li>
<li>✅ Session timeout: 10 minutes</li>
<li>✅ All pages protected by session check</li>
</ul>

<form method='post' action='logout' style='display: inline;'>
<button type='submit'>Logout</button>
</form>

<br><br>
<a href='login'>Login Page</a> |
<a href='profile'>Profile Page</a>
</div>
</body>
</html>
```

OUTPUT:





RESULT

A Java Servlet application was successfully implemented to demonstrate session tracking using `HttpSession`.